1.47 Proof by Induction: Divisibility

Worked Solutions

1 a)

$$(2n+1)(2n-3)-4n+23 = 4n^2-4n-3-4n+23$$
$$= 4n^2-8n+20 = 4(n^2-2n+5)$$

$\therefore (2n+1)(2n-3)-4n+23$ is divisible by 4.

1 b)

$$16^{3k}-4 = (4^2)^{3k}-4 = 4^{6k}-4 = 4(4^{6k-1}-1)$$

$\therefore 16^{3k}-4$ is divisible by 4.

1 c)

$$2^{k+2}-4\times 3^k+8 = 2^2\times 2^k-4\times 3^k+8 = 4(2^k-3^k+2)$$

$\therefore 2^{k+2}-4\times 3^k+8$ is divisible by 4.

2 a)

$$6^{k+1}-18 = 6\times 6^k-18 = 6(6^k-3)$$

$\therefore 6^{k+1}-18$ is divisible by 6.

2 b)

$$2^{k+1}\times 3^{k+2} = 2\times 2^k\times 3\times 3^{k+1} = 6(2^k\times 3^{k+1})$$

$\therefore 2^{k+1}\times 3^{k+2}$ is divisible by 6.

2 c)

$$(k^2+1)(k-4)+10(k^2+k+1) = k^3-4k^2+k-4+10k^2+10k+10$$
$$= k^3+6k^2+11k+6 = (k+1)(k+2)(k+3),$$

which is 3 consecutive integers. At least one of these factors will be even so the expression is a multiple of 2. One of the factors will be a multiple of 3. As the expression is a multiple of both 2 and 3, it must be a multiple of 6. $\therefore (k^2+1)(k-4)+10(k^2+k+1)$ is divisible by 6.

3 a)

False: if $a=2$, $b=3$ and $c=6$ then $a+b=5$ which does not divide 6.

3 b)

True:
Let $pa=b$ and $qa=c$ for some values p and q. Then $b+c=pa+qa=a(p+q)$ so a is a factor of $b+c$.

3 c)

False: if $a = 5$, $b = 2$ and $c = 3$ then $b + c = 5$ so a is a factor of $b + c$ but b and c are not factors of a.

3 d)

True:
Let $pa = b$ and $qb = c$ for some values p and q. Then $b + c = pa + qb = pa + pqa = a(p + pq)$ so a is a factor of $b + c$.

4)

Prove true for $n = 1$:

$$1 + 1 = 1^2 \text{ which is divisible by 2 so true for } n = 1.$$

Assume true for $n = k$: $k^2 + k$ is divisible by 2.

Prove true for $n = k + 1$:

$$(k + 1)^2 + (k + 1) = k^2 + 2k + 1 + k + 1$$

$$= k^2 + 3k + 2 = (k^2 + k) + 2k + 2 = (k^2 + k) + 2(k + 1).$$

Since we assumed $k^2 + k$ is divisible by 2 and $2(k + 1)$ is divisible by 2, $(k^2 + k) + 2(k + 1)$ must be divisible by 2.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

5 a)

Prove true for $n = 1$: $1 - 1 = 0$ which is divisible by 3 so true for $n = 1$.

Assume true for $n = k$: $k^3 - k$ is divisible by 3.

Prove true for $n = k + 1$:

$$(k + 1)^3 - (k + 1) = k^3 + 3k^2 + 3k + 1 - k - 1$$

$$= k^3 + 3k^2 + 2k = (k^3 - k) + 3k^2 + 3k = (k^3 - k) + 3(k^2 + k).$$

Since we assumed $k^3 - k$ is divisible by 3 and $3(k^2 + k)$ is divisible by 3, $(k^3 - k) + 3(k^2 + k)$ must be divisible by 3.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

5 b)

Prove true for $n = 1$: $1 + 3 + 2 = 6$ which is divisible by 6 so true for $n = 1$.

Assume true for $n = k$: $k^3 + 3k^2 + 2k$ is divisible by 6.

Prove true for $n = k + 1$:

$$(k + 1)^3 + 3(k + 1)^2 + 2(k + 1)$$

$$= k^3 + 3k^2 + 3k + 1 + 3k^2 + 6k + 3 + 2k + 2$$

$$= k^3 + 6k^2 + 11k + 6 = (k^3 + 3k^2 + 2k) + 3k^2 + 9k + 6$$

$$= (k^3 - k) + 3(k^2 + 3k + 2) = (k^3 - k) + 3(k + 1)(k + 2).$$

Since we assumed $k^3 - k$ is divisible by 6, $(k+1)$ and $(k+2)$ are consecutive integers so one must be even so that $3(k+1)(k+2)$ must be divisible by both 2 and 3 therefore divisible by 6. $(k^3 - k) + 3(k+1)(k+2)$ must be divisible by 6.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

6)

Prove true for $n = 1$: $9^1 + 7 = 16$ which is divisible by 8 so true for $n = 1$.

Assume true for $n = k$: $9^k + 7$ is divisible by 8.

Prove true for $n = k + 1$:

$$9^{k+1} + 7 = 9 \times 9^k + 63 - 56 = 9(9^k + 7) - 56.$$

Since we assumed $9^k + 7$ is divisible by 8 and 56 is divisible by 8 then $9(9^k + 7) - 56$ must be divisible by 8.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

7)

Prove true for $n = 1$: $5^1 - 4 + 3 = 4$ which is divisible by 4 so true for $n = 1$.

Assume true for $n = k$: $5^k - 4k + 3$ is divisible by 4.

Prove true for $n = k + 1$:

$$5^{k+1} - 4(k+1) + 3 = 5 \times 5^k - 4k - 1$$

$$= 5 \times 5^k - 20k + 15 + 16k - 16 = 5(5^k - 4k + 3) + 4(4k - 4).$$

Since we assumed $5^k - 4k + 3$ is divisible by 4 and $4(4k - 4)$ is divisible by 4 then $5(5^k - 4k + 3) + 4(4k - 4)$ must be divisible by 4.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

8)

Prove true for $n = 1$: $5^1 + 4 - 1 = 8$ which is divisible by 8 so true for $n = 1$.

Assume true for $n = k$: $5^k + 4k - 1$ is divisible by 8.

Prove true for $n = k + 1$:

$$5^{k+1} + 4(k+1) - 1 = 5 \times 5^k + 4k + 3$$

$$= 5 \times 5^k + 20k - 5 - 16k + 8 = 5(5^k + 4k - 1) - 8(2k - 1).$$

Since we assumed $5^k + 4k - 1$ is divisible by 8 and $8(2k - 1)$ is divisible by 8 then $5(5^k + 4k - 1) - 8(2k - 1)$ must be divisible by 8.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

9)

Prove true for $n = 1$: $6^1 - 2^1 = 4$ which is a multiple of 4 so true for $n = 1$.

Assume true for $n = k$: $6^k - 2^k$ is a multiple of 4.

Prove true for $n = k + 1$:

$$6^{k+1} - 2^{k+1} = 6 \times 6^k - 2 \times 2^k$$

$$= 2 \times 6^k + 4 \times 6^k - 2 \times 2^k = 2(6^k - 2^k) + 4 \times 6^k.$$

Since we assumed $6^k - 2^k$ is a multiple of 4 and 4×6^k is a multiple of 4 then $2(6^k - 2^k) + 4 \times 6^k$ must be a multiple of 4.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

10 a)

Prove true for $n = 1$: $5^1 + 3 \times 9^1 = 32$ which is divisible by 4 so true for $n = 1$.

Assume true for $n = k$: $5^k + 3 \times 9^k$ is divisible by 4.

Prove true for $n = k + 1$:

$$5^{k+1} + 3 \times 9^{k+1} = 5 \times 5^k + 3 \times 9 \times 9^k$$

$$= 5 \times 5^k + 27 \times 9^k = 5 \times 5^k + 15 \times 9^k + 12 \times 9^k$$

$$= 5(5^k + 3 \times 9^k) + 12 \times 9^k.$$

Since we assumed $5^k + 3 \times 9^k$ is divisible by 4 and 12×9^k is divisible by 4 then $5(5^k + 3 \times 9^k) + 12 \times 9^k$ must be divisible by 4.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

10 b)

Prove true for $n = 1$: $5^2 + 3^{2-1} = 28$ which is divisible by 4 so true for $n = 1$.

Assume true for $n = k$: $5^{2k} + 3^{2k-1}$ is divisible by 4.

Prove true for $n = k + 1$:

$$5^{2(k+1)} + 3^{2(k+1)-1} = 5^{2k+2} + 3^{2k+1}$$

$$= 25 \times 5^{2k} + 3^2 \times 3^{2k-1} = 25 \times 5^{2k} + 9 \times 3^{2k-1}$$

$$= 25 \times 5^{2k} + 25 \times 3^{2k-1} - 16 \times 3^{2k-1}$$

$$= 25(5^{2k} + 3^{2k-1}) - 16 \times 3^{2k-1}.$$

Since we assumed $5^{2k} + 3^{2k-1}$ is divisible by 4 and $16 \times 3^{2k-1}$ is divisible by 4 then $25(5^{2k} + 3^{2k-1}) - 16 \times 3^{2k-1}$ must be divisible by 4.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

11)

Prove true for $n = 2$: $15^2 - 8^{2-2} = 224$ which is divisible by 7 so true for $n = 2$.

Assume true for $n = k$: $15^k - 8^{k-2}$ is divisible by 7.

Prove true for $n = k + 1$:

$$15^{k+1} - 8^{k+1-2} = 15 \times 15^k - 8 \times 8^{k-2}$$

$$= 15 \times 15^k - 15 \times 8^{k-2} + 7 \times 8^{k-2}$$

$$= 15(15^k - 8^{k-2}) + 7 \times 8^{k-2}.$$

Since we assumed $15^k - 8^{k-2}$ is divisible by 7 and $7 \times 8^{k-2}$ is divisible by 7 then $15(15^k - 8^{k-2}) + 7 \times 8^{k-2}$ must be divisible by 7.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

12)

Prove true for $n = 1$: $3^1 + 7^1 = 10$ which is divisible by 10 so true for $n = 1$.

Assume true for $n = k$ where k is odd: $3^k + 7^k$ is divisible by 10.

Prove true for $n = k + 2$:

$$3^{k+2} + 7^{k+2} = 3^2 \times 3^k + 7^2 \times 7^k$$
$$= 9(3^k + 7^k) + 40 \times 7^k.$$

Since we assumed $3^k + 7^k$ is divisible by 10 and 40×7^k is divisible by 10 then $9(3^k + 7^k) + 40 \times 7^k$ must be divisible by 10.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ where k is odd, then the statement is true for $n = k + 2$, so the statement is true for all odd integers $n \geq 1$.

13 a)

Errors:
Didn't check true for $n = 1$.
Didn't state "assume true for $n = k$".
Only attempted to show that the difference in two terms is divisible by 8 rather than specifically looking at the $k + 1$ case.

13 b)

Let $f(k) = 3^{2k} - 1$.

Prove true for $n = 1$: $3^2 - 1 = 8$ which is a multiple of 8 so true for $n = 1$.

Assume true for $n = k$: $3^{2k} - 1$ is a multiple of 8.

Prove true for $n = k + 1$:

$$f(k+1) - f(k) = (3^{2(k+1)} - 1) - (3^{2k} - 1)$$
$$f(k+1) - 8m = 3^2 \times 3^{2k} - 3^{2k} \quad \text{where } m \text{ is some integer}$$
$$f(k+1) = 8m + (9-1) \times 3^{2k}$$
$$f(k+1) = 8m + 8 \times 3^{2k} = 8[m + 3^{2k}],$$

which is divisible by 8.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

14)

Let three consecutive positive integers be n, $n+1$ and $n+2$ therefore proving $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 3.

Prove true for $n = 1$: $1^3 + 2^3 + 3^3 = 36$ which is divisible by 3 so true for $n = 1$.

Assume true for $n = k$: $k^3 + (k+1)^3 + (k+2)^3$ is divisible by 3.

Prove true for $n = k + 1$:

$$(k+1)^3 + (k+2)^3 + (k+3)^3$$
$$= k^3 + 3k^2 + 3k + 1 + k^3 + 6k^2 + 12k + 8 + k^3 + 9k^2 + 27k + 27$$
$$= 3k^3 + 18k^2 + 42k + 36 = 3[k^3 + 6k^2 + 14k + 12],$$

which is divisible by 3.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

15 a)

$$f(n+1) - f(n) = 50(n+1)^2 + 70(n+1) - (50n^2 + 70n)$$
$$= 50n^2 + 100n + 50 + 70n + 70 - 50n^2 - 70n$$
$$= 100n + 120 = 20(5n+6) \text{ as req.}$$

15 b)

Prove true for $n = 1$: $f(1) = 50 + 70 = 120$ which is divisible by 4 so true for $n = 1$.

Assume true for $n = k$: so let $f(k) = 4m$ for some integer m.

Prove true for $n = k + 1$:

$$f(k+1) - f(k) = 20(5k+6)$$
$$f(k+1) - 4m = 20(5k+6)$$
$$f(k+1) = 4m + 20(5k+6) = 4[m + 5(5k+6)],$$

which is divisible by 4.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

16 a)

$$f(n+1) - f(n) = 24 \times 2^{4(n+1)} + 3^{4(n+1)} - 24 \times 2^{4n} - 3^{4n}$$
$$= 24 \times 2^{4n+4} + 3^{4n+4} - 24 \times 2^{4n} - 3^{4n}$$
$$= 384 \times 2^{4n} + 81 \times 3^{4n} - 24 \times 2^{4n} - 3^{4n}$$
$$= 360 \times 2^{4n} + 80 \times 3^{4n}.$$

16 b)

Prove true for $n = 1$: $f(1) = 24 \times 2^4 + 3^4 = 465$ which is divisible by 5 so true for $n = 1$.

Assume true for $n = k$: so let $f(k) = 5m$ for some integer m.

Prove true for $n = k + 1$:

$$f(k+1) - f(k) = 360 \times 2^{4k} + 80 \times 3^{4k}$$
$$f(k+1) - 5m = 360 \times 2^{4k} + 80 \times 3^{4k}$$
$$f(k+1) = 5m + 360 \times 2^{4k} + 80 \times 3^{4k}$$
$$f(k+1) = 5[m + 72 \times 2^{4k} + 16 \times 3^{4k}],$$

which is divisible by 5.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

17 a)

Prove true for $n = 1$: $2 - 5 = -3$ which is a multiple of 3 so true for $n = 1$.

Assume true for $n = k$: $2k^3 - 5k$ is a multiple of 3.

Prove true for $n = k + 1$:

$$2(k+1)^3 - 5(k+1) = 2k^3 + 6k^2 + 6k + 2 - 5k - 5$$

$$= 2k^3 + 6k^2 + k - 3 = (2k^3 - 5k) + 6k^2 + 6k - 3$$

$$= (2k^3 - 5k) + 3(2k^2 + 2k - 1).$$

Since we assumed $2k^3 - 5k$ is a multiple of 3 and $3(2k^2 + 2k - 1)$ is a multiple of 3, $(2k^3 - 5k) + 3(2k^2 + 2k - 1)$ must be a multiple of 3.

As the statement is true for $n = 1$, and if the statement is true for $n = k$ then the statement is true for $n = k + 1$, so the statement is true for all integers $n \geq 1$.

17 b)

$$f(-x) = 2(-x)^3 - 5(-x) = -2x^3 + 5x = -(2x^3 - 5x) = -f(x).$$

17 c)

From part a), $f(n)$ is a multiple of 3 for all integers $n \geq 1$. From part b), $f(-n) = -f(n)$ so is a multiple of 3 so is true for all integers $n \leq 1$. $f(0) = 0$ which is a multiple of 3. $\therefore f(n)$ is a multiple of 3 for all integer values of n.